

Jiaming Shan

jiamingshan@ucsb.edu | shanjiaming.github.io

Research Interests

Core Interests: LLM Efficiency, Model Architecture, Speculative Decoding, Sparse & Long-Context Attention, Autonomous Agents

Education

Shanghai Jiao Tong University

Bachelor of Computer Science

Sep 2020 – Jun 2024

- **Member of ACM Honors Class, an elite CS program for top 5% talented students**
- **GPA (All Core Courses): 3.98/4.3, Ranking: 3/36**
- **2023 National Scholarship Award**

University of California, Santa Barbara

PhD student in Computer Science

Sep 2024 – Present

Advisor: Prof. Xifeng Yan

Research Experience

University of California, Santa Barbara

PhD Student

Sep 2024 – Present

Advisor: Prof. Xifeng Yan

- **[Learning When Not to Attend Globally](#)** (All-or-Here Attention, AHA). Jiaming Shan*, Xuan Luo*, Wesley Truong, Kailai Zhang, Hanzhe Zhang, Xifeng Yan. *Under review for EMNLP 2026 via ACL ARR May 2026*. Proposed a token-level adaptive attention mechanism that equips each attention head with a binary router to switch between full attention (“All”) and local sliding-window attention (“Here”). (* Equal Contribution) Contributed to real-world acceleration implementation and inference-speedup code paths; experiments show AHA can replace up to 88% of full-attention operations with sliding-window attention without performance loss, and can complement sparse-attention frameworks such as DuoAttention to further reduce global-attention overhead on long-context tasks.
- **DFlash Exploration.** Explored end-to-end fine-tuning for DFlash acceleration.
- **[Proactive Agent Research Environment: Simulating Active Users to Evaluate Proactive Assistants](#)**. Deepak Nathani, Cheng Zhang, Chang Huan, Jiaming Shan, Yinfei Yang, Alkesh Patel, Zhe Gan, William Yang Wang, Michael Saxon, Xin Eric Wang. *Under review at COLM 2026; arXiv preprint, 2026*. Built benchmark and environment abstractions for proactive and mobile agents, including FSM-based app modeling, goal inference, and multi-app orchestration evaluation.

Massachusetts Institute of Technology (MIT)

Summer Research, CSAIL

Aug 2023 – Jun 2024

Advisor: Prof. Chuang Gan

- **[Building Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge](#)**. Weihua Du, Qiushi Lyu, Jiaming Shan, Zhenting Qi, Hongxin Zhang, Sunli Chen, Andi Peng, Tianmin Shu, Kwonjoon Lee, Behzad Dariush, Chuang Gan. *NeurIPS 2024, Datasets and Benchmarks Track. (Poster)* Conceptualized and implemented physically constrained agents and helper bots within the ThreeDWorld (TDW) simulation environment; leveraged in-context learning to deploy language-based agents for human assistance tasks and generate a new benchmark dataset.

MIT / Shanghai Jiao Tong University

Remote Research

Feb 2023 – May 2023

Advisor: Prof. Chuang Gan

- **[Building Cooperative Embodied Agents Modularly with Language Models](#)**. Hongxin Zhang*, Weihua Du*, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan. *ICLR 2024. (Poster)* (* Equal Contribution) Co-developed a prompt-based state abstraction framework for multi-agent cooperation. Led VirtualHome environment and heuristic-baseline implementation, and conducted user studies comparing LLM agents with planning baselines.

Internship Experience

RWKV

Mar 2026 – Jun 2026

Agent Intern

Built agent harnesses around RWKV’s local deployment and high-concurrency strengths:

- Developed local high-concurrency document-processing pipelines, including document summarization and document retrieval.
- Used RWKV to summarize tool outputs and extract key information, reducing downstream token consumption.
- Concurrently generated HTML views after producing the main output, giving users a more readable presentation layer.

Agent Daily Use

Auto Research in Scientific Work: Explored using auto-research workflows in day-to-day research, including question decomposition, reference search and synthesis, experiment planning, and turning intermediate reasoning into reusable research notes; summarized practical lessons in [notes](#).

TA Grading Agent: Converted the Gradescope grading interface into MCP tools and agent workflows for batch homework grading; used dual-pass grading plus human verification queues to protect correctness, reducing TA grading workload by roughly 70%.

Thought-to-Blog Agent: Built a pipeline that turns high-signal AI chat records into publishable blog drafts by capturing native human thought streams, screening privacy/value risk, and generating working/final/published versions; addresses both hollow AI-generated writing and the fatigue of manual blogging.

Autonomous Wealth Research Agent: Built a daily investing automation connected to moomoo and a local agent workforce; generates portfolio-aware recommendations using supply-chain research, market context, tax/cash-flow constraints, risk gates, and simulated trading for autonomous experimentation.

OpenClaw-like Personal Agent Framework: Developed and daily-used a personal agent framework before mainstream CoWork-style releases, integrating memory, sleep/compaction, MCP tool routing, and long-running personal workflows.

Honors & Awards

- 2023 National Scholarship Award
- 2020, 2021, 2022, 2023 Zhiyuan Honorary Scholarship (Top 2% in Shanghai Jiao Tong University)
- 2021 Interdisciplinary Contest In Modeling Honorable Mention
- 2021 Third Prize, National Undergraduate Mathematical Contest in Modeling, Provincial Level
- 2021 Silver Medal, International Physics Competition for University Students

Technical Skills

AI / ML Frameworks:	PyTorch, Hugging Face, Triton, FlashAttention, vLLM, Weights & Biases
AI-Native Workflow:	Codex, OpenClaw, Cursor, Claude Code, Tmux
Programming Languages:	Python, C++, Lisp, Java
Developer Tools & MLOps:	Git, Docker, LaTeX

单佳铭 (Jiaming Shan)

✉ jiamingshan@ucsb.edu | 🌐 shanjiaming.github.io | 📍 美国加利福尼亚州圣巴巴拉

研究方向

核心兴趣：LLM 推理效率、模型架构、投机解码、稀疏与长上下文注意力、自主智能体

教育背景

上海交通大学 (SJTU)

2020.09 – 2024.06

计算机科学学士 (ACM 班)

上海, 中国

- 入选 ACM 班 (全系前 5% 选拔出的精英计算机试点班)
- 绩点 (核心课程): **3.98/4.3**, 排名: **3/36**
- **2023 年国家奖学金**

加州大学圣塔芭芭拉分校 (UCSB)

2024.09 – 至今

计算机科学博士在读

导师: Prof. Xifeng Yan

科研经历

加州大学圣塔芭芭拉分校 (UCSB)

2024.09 – 至今

计算机科学博士在读

导师: Prof. Xifeng Yan

- **Learning When Not to Attend Globally** (All-or-Here Attention, AHA). Jiaming Shan*, Xuan Luo*, Wesley Truong, Kailai Zhang, Hanzhe Zhang, Xifeng Yan. *Under review for EMNLP 2026 via ACL ARR May 2026*. 提出 token-level adaptive attention 机制, 为每个 attention head 配置二元 router, 在全局注意力 (“All”) 与局部滑动窗口注意力 (“Here”) 间动态切换。(* 共同一作)
参与真实推理加速实现与 inference-speedup code paths; 实验显示 AHA 可用 sliding-window attention 替代最多 88% full-attention operations 且无性能损失, 并可与 DuoAttention 等稀疏注意力框架互补, 进一步降低长上下文任务中的 global-attention overhead.
- **DFlash 相关探索**. 尝试通过端到端微调加速 DFlash.
- **Proactive Agent Research Environment: Simulating Active Users to Evaluate Proactive Assistants**. Deepak Nathani, Cheng Zhang, Chang Huan, Jiaming Shan, Yinfei Yang, Alkesh Patel, Zhe Gan, William Yang Wang, Michael Saxon, Xin Eric Wang. *Under review at COLM 2026; arXiv preprint, 2026*.
面向主动型与移动端智能体, 构建基准与环境抽象, 包括基于有限状态机的应用建模、意图推理和多应用调度评估。

麻省理工学院 (MIT)

2023.08 – 2024.06

暑期研究 (CSAIL)

波士顿, 美国

- **Building Constrained Human-AI Cooperation: An Inclusive Embodied Social Intelligence Challenge**. Weihua Du, Qiushi Lyu, Jiaming Shan, Zhenting Qi, Hongxin Zhang, Sunli Chen, Andi Peng, Tianmin Shu, Kwonjoon Lee, Behzad Dariush, Chuang Gan. *NeurIPS 2024 (Datasets and Benchmarks Track)*. (Poster)
在 ThreeDWorld (TDW) 环境中设计并实现物理受限的协作智能体与 helper bots; 利用 LLM 的 in-context learning 部署基于语言的协作机器人, 并构建评测基准。

MIT / 上海交通大学

2023.02 – 2023.05

远程研究

导师: 涂创教授

- **Building Cooperative Embodied Agents Modularly with Language Models**. Hongxin Zhang*, Weihua Du*, Jiaming Shan, Qinghong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, Chuang Gan. *ICLR 2024*. (Poster)
(* 共同一作)
合作开发将环境状态转换为文本提示的协作框架; 参与 VirtualHome 环境与启发式 baseline 实现, 并通过用户研究比较 LLM 智能体与传统规划方法。

实习经历

RWKV

2026.03 – 2026.06

Agent 方向实习生

- 利用 RWKV 的本地部署与高并发特性构建 agent harness:
- 实现本地高并发文档处理能力，包括文档概括与文档检索。
 - 使用 RWKV 对工具输出进行 summary 和关键信息提取，减少后续主模型 token 消耗。
 - 在产生主输出结果后，并发生成 HTML 展示页面，提升用户侧可读性与结果呈现质量。

Agent 日常落地

科研 Auto Research 探索: 探索在科研中使用 auto research，将问题拆解、文献检索与综合、实验构想和阶段性推理沉淀为可持续迭代的研究 workflow，并总结出一些实践经验：[实践经验总结](#)。

TA 批改 Agent: 将 Gradescope 批改网页界面封装为 MCP 工具与 agent workflow，用于批量作业批改；通过双重批改与人工验证队列保障正确性，节省约 70% TA 批改工作量。

思考流到博客 Agent: 构建从高质量 AI 聊天记录到可发布博客的自动化流水线，捕获人类原生思考流，进行隐私/价值筛选，并生成工作稿、最终稿与发布版，缓解 AI 生成内容空洞与人工写作疲劳。

自动投资研究 Agent: 构建接入 moomoo 与本地 automation 的每日投资研究系统，根据当日仓位、供应链深度研究、市场环境、税务/现金流约束和风险门控生成调仓建议，并维护模拟交易体系供 AI 自主试验。

OpenClaw 式个人 Agent 框架: 在 mainstream CoWork-style releases 前自建并日常使用个人 agent 框架，整合 memory、sleep/compaction、MCP tool routing 与长期个人 workflow。

荣誉奖项

- 2023 年国家奖学金 (本科生最高荣誉)
- 2020-2023 年上海交通大学“致远荣誉奖学金”(连续四年获得, Top 2%)
- 2021 年国际大学生数学建模竞赛 (MCM/ICM) Honorable Mention
- 2021 年国际大学生物理竞赛 (IPC) 银奖

专业技能

人工智能框架:	PyTorch, Hugging Face, Triton, FlashAttention, vLLM, Weights & Biases
AI 原生工作流:	Codex, OpenClaw, Cursor, Claude Code, Tmux
编程语言:	Python, C++, Lisp, Java
工具与部署:	Git, Docker, LaTeX